CCS2131 DATA STRUCTURES AND ALGORITHMS

SEMESTER DEFERRED EXAMINATION

FRIDAY, 19 JUNE 2026

14:00 – 17:00 HOURS

TIME ALLOWED : WRITING — THREE HOURS : READING —5 MINUTES

1. **Answer *all* parts of Section A (40 marks).**
2. **Section B (60 marks)** – answer **any *three* questions** only.
3. Begin each question on a new page.
4. Show *all* working; marks are awarded for method as well as the final answer.

SECTION A: COMPULSORY — 40 MARKS

Question 1

(a) A university library system stores book records using different data structures.

1. Explain the difference between **linear** and **non-linear** data structures, giving two examples of each. (4)
2. Distinguish between **static** and **dynamic** data structures. (3)
3. State three reasons why a programmer may choose a linked list instead of an array. (3)

(10 marks)

(b) A stack is used to check whether brackets in an expression are correctly balanced.

Given the expression:

```
{ A + (B * C) - [D / (E + F)] }
```

1. Show the sequence of stack operations used to process the brackets. (6)
2. State whether the expression is balanced or not. (2)
3. Mention two common applications of stacks in computing. (2)

(10 marks)

(c) A queue is used in a printer system where documents are printed in the order they arrive.

1. Explain how a **circular queue** solves the problem of wasted space in a normal linear queue. (4)
2. Using a circular queue of size 6, insert the following jobs:

   `J1, J2, J3, J4`

   Then delete two jobs and insert:

   `J5, J6, J7`

   Show the contents of the queue after each major operation. (4)
3. State two conditions that must be checked before performing queue operations. (2)

(10 marks)

(d) A Binary Search Tree is created by inserting the following values in order:

`45, 25, 60, 15, 30, 50, 70, 28`

1. Draw the final Binary Search Tree. (4)
2. Show the result of the following traversals:
   - In-order traversal
   - Pre-order traversal
   - Post-order traversal (3)

3. Explain why the in-order traversal of a BST is useful. (3)

(10 marks)

SECTION B: ATTEMPT ANY THREE QUESTIONS — 60 MARKS

Question 2

(a) A singly linked list contains student numbers arranged as follows:

`102 → 115 → 128 → 140 → NULL`

Write an algorithm or pseudo-code to insert a new student number `121` after the node containing `115`. Clearly show the pointer changes required.

(10 marks)

(b) Hash the following keys into a table of size 13 using **linear probing**:

`18, 31, 44, 57, 22, 35, 48`

Use the hash function:

`h(k) = k mod 13`

Show:

1. The initial hash position for each key.
2. Any collision that occurs.
3. The final hash table.

(10 marks)

Question 3

(a) Write a recursive function `power(base, exponent)` that calculates:

`base^exponent`

For example:

`power(3, 4) = 81`

Your answer should include:

1. The base case. (3)
2. The recursive case. (3)
3. A trace of the function calls for `power(2, 5)`. (4)

(10 marks)

(b) Use **Insertion Sort** to arrange the following numbers in ascending order:

`34, 12, 45, 9, 28, 16`

Show the list after each pass.

(10 marks)

Question 4

(a) A skip list is used to improve search performance in a sorted list.

1. Describe the structure of a skip list. (4)
2. Explain how searching is performed in a skip list. (4)
3. Give two advantages of skip lists over ordinary linked lists. (2)

(10 marks)

(b) Insert the following keys into an initially empty **min-heap**:

`40, 18, 25, 10, 30, 5, 12`

Show the heap after each insertion and draw the final heap as a binary tree.

(10 marks)

Question 5

(a) A graph has the following adjacency list:

- A: B, C
- B: D, E
- C: F
- D: G
- E: G
- F: E
- G: none

1. Perform a **Breadth-First Search** starting from vertex A. (5)
2. Perform a **Depth-First Search** starting from vertex A. (5)

(10 marks)

(b) Using the following weighted graph edges:

- A–B = 4
- A–C = 2
- B–C = 1
- B–D = 5
- C–D = 8
- C–E = 10
- D–E = 2

- D–F = 6
- E–F = 3

Apply **Dijkstra's Algorithm** starting from vertex A.

Show:

1. The tentative distances after each major step.
2. The predecessor of each vertex.
3. The shortest path from A to F.

(10 marks)

END OF EXAMINATION